

Social Media Engagement Analytics — Summary of Findings

Module End Assignment · Dataset: social_media_engagement_5000.csv (5,000 posts, 21 columns)

Data Cleaning

age, gender, likes, comments, shares and **sentiment** each had 150 missing values (3%). Numeric columns were filled with the median, categorical columns with the mode; no rows were dropped. No duplicate rows were found. About 12% of posts (603 of 5,000) had an engagement rate above 100% (max ≈ 191.5), since likes/comments/shares exceeded impressions for low-impression posts — these were capped, giving **engagement_rate_clean** (0–1), which all analysis below uses.

Key Findings

- **Content performance is very flat.** Engagement rate differs by only 1–4 percentage points across post type (text 0.376, video 0.376, image 0.374, reel 0.360 — lowest) and category (fitness highest at 0.384, food lowest at 0.357). No format or category is a clear winner or loser.
- **Country has the widest spread.** Australia has the highest average engagement rate (0.395), followed by Japan (0.385) and UAE (0.383); India is lowest (0.340) — a real but modest ~5-point gap.
- **Age barely matters.** Correlation between age and engagement rate is essentially zero ($r = -0.02$). The youngest group (13–17) engages slightly more (0.398) than the oldest (55–64, 0.356), but this is a small effect, not a strong age trend.
- **Verified accounts underperform slightly.** Verified accounts have a lower average engagement rate (0.349) than unverified ones (0.374) despite similar impressions (~50,000 vs ~49,900) and fewer average likes (9,051 vs 9,501) — verification does not translate into higher engagement here.
- **Weak timing signal.** Tuesday (avg 51,309 impressions) and Saturday (50,945) post slightly higher than Thursday (48,119, lowest); the dataset has only a date, not a clock time, so "time of day" could not be analysed directly.
- **Device has almost no effect on watch time** (mobile 4,088s, tablet 3,980s, desktop 3,975s — a ~3% spread).
- **Sentiment does not predict performance.** Positive posts are 50% of the data, neutral 31%, negative 19%. Engagement rate is nearly identical across all three (0.368–0.375), and negative posts perform on par with positive ones rather than underperforming.
- **Impressions drive apparent engagement rate** most strongly of any numeric relationship ($r = -0.78$ with **engagement_rate_clean**): low-impression posts mechanically show higher rates, which is why the capping step in cleaning matters for fair comparison.

Analysis & Visualizations Performed

Stage	Method / Chart	Purpose
Exploration	describe(), value_counts(), groupby()	Distributions, categorical counts, averages by group
Wrangling	merge, concat, engagement_score, log features	New metrics; country-level stats joined back
Statistics	mean/median/mode, std, percentiles, skew/kurtosis	Likes, comments, shares, watch time, engagement, followers
Matplotlib	Scatter, line, bar, pie, histogram, box (6 plots)	Likes vs impressions, daily trend, category counts, gender split, age, engagement spread
Seaborn	Count, bar, violin, pair, heatmap, swarm (6 plots)	Post-type counts, category averages, sentiment distributions, correlations
Plotly	Interactive line, bar, bubble, scatter (4 plots)	Monthly trend, country comparison, category bubble, likes vs impressions by type

Overall takeaway: engagement on this dataset is largely uniform across content type, category, device and sentiment — the only meaningfully sized effects are by country and impression volume, suggesting reach/targeting matters more than content format or tone for this platform's data.